Aluffi 1-5 solutions

pika

July 22, 2026

Exercise. [Aluffi, Algebra, Ch-1, Ex-5.1] Prove that a final object in a category C is initial in the opposite category C^{op}.

Solution. Suppose I is the final object in C. Then, take any object $Z \in \mathsf{C}^{\mathsf{op}}$. Since there is always a unique morphism $Z \to I$ in C, there is a unique morphism $I \to Z$ in C^{op}. Siince $Z \in \mathsf{C}^{\mathsf{op}}$ was arbitrary, this proves that I is the initial object in C^{op}. ■

Exercise. [Aluffi, Algebra, Ch-1, Ex-5.2] Prove that $\varnothing$ is the *unique* initial object in Set.

Solution. Suppose $X \in \mathsf{Obj}(\mathsf{Set})$ is an initial object. Then, there is a unique morphism $X \to \varnothing$ which proves that $X = \varnothing$. ■

Exercise. [Aluffi, Algebra, Ch-1, Ex-5.3] Prove that final objects are unique up to isomorphisms.

Solution. Suppose there are two final objects F_1, F_2 in a category C. Then, since F_2 is final, we pick the unique morphism $f : F_1 \to F_2$ and since F_1 is final we consider the unique morphism $g : F_2 \to F_1$. Then, notice that $fg = 1_{F_2}$. This is because F_2 is final, so there is only a unique morphism $1_{F_2} : F_2 \to F_2$ and gf must be identical to this. By a similar argument we deduce that $gf = 1_{F_1}$. This completes the proof that any two final objects F_1 and F_2 are always isomorphic. ■

Exercise. [Aluffi, Algebra, Ch-1, Ex-5.4] What are the final objects in the category of 'pointed sets' in Eg-3.8? Are they unique?

Solution. Any singleton $\{*\}$ with the point $*$ is a final object. Indeed, given any object (Z, z) in Set^* there is obviusly a morphism $\sigma : Z \to \{*\}$ so that $\sigma(z) = *$. This is clearly the constant map $Z \to \{*\}$. Any other singleton set S with its point s forming the pair (S, s) would be a final object in this category so such a construction is not unique although obviosuly by a previous exercise it would be unique up to isomorphism. ■

Exercise. [Aluffi, Algebra, Ch-1, Ex-5.5] What are the final objects in the category considered in Eg-5.3?

Solution. The final object should be $(c : a \mapsto *, \{*\})$. Indeed, in this way, given any object (f, Z) where f agrees on equivalent points of A, we get a morphism $\sigma : (f, Z) \to (c, \{*\})$ that is constant in Z (maps every point in Z to $*$). Such a morphism is guaranteed to be unique. ■

Exercise. [Aluffi, Algebra, Ch-1, Ex-5.6] Consider the category corresponding to Eg-3.3 by endowing the set $\mathbb{Z}_{>0}$ of positive integers with the *divisibility relation*. Thus, there is exactly one morphism $d \to m$ in this category if and only if d divides m without remainder; there is no morphism between d and m otherwise. Show that this category has products and coproducts. What are their conventional names?

Solution. That this construction defines a category is obvious so we will not bother with proving that part. Notice that the $\gcd(a, b)$ divides both a and b so there are morphisms $\gcd(a, b) \to a, \gcd(a, b) \to b$. Next, given any other integer z with pairs of morphisms $z \to a$ and $z \to b$, there is a (unique by construction of this category) morhpism $z \to \gcd(a, b)$ simply by definition of gcd(a,b). ■

Exercise. [Aluffi, Algebra, Ch-1, Ex-5.8] Show that in every category C the products $A \times B$ and $B \times A$ are isomorphic, if they exist.

Solution. Suppose C is a category with objects A, B so that the products $A \times B$ and $B \times A$ both exist. We denote the canonical projections by $p_A : A \times B \to A, p_B : A \times B \to B, p'_B : B \times A \to B, p'_A : B \times A \to A$. Next, by the universal property of $A \times B$, we get a unique morphism $f : B \times A \to A \times B$ so that $p_A f = p'_A, p_B f = p'_B$. Next, by the universal property of $B \times A$, there is a unique morphism $g : A \times B \to B \times A$ so that

$$p'_B g = p_B, p'_A g = p_A$$

Now, we see that $p_A f = p'_A gf = p'_A$ and $p_B f = p'_B gf = p'_B$. Also, we see that $p'_B g = p_B fg = p_B$ and $p'_A g = p_A fg = p_A$. Notice that $1_{B\times A}$ is a morphism $B \times A \to B \times A$ that satisfies the first two relations the same way as gf. By uniqueness, we get that $gf = 1_{B\times A}$. Similarly, $1_{A\times B}$ is a morphism $A \times B \to A \times B$ that satisfies the third and the fourth commutativity relations above the same way as fg. Once again, by uniqueness we find that $fg = 1_{A\times B}$. This completes the proof. ■

Exercise. [Aluffi, Algebra, Ch-1, Ex-5.9] Let C be a category with products. Find a reasonable candidate for the universal property that the product $A \times B \times C$ of three objects of C ought to satisfy, and prove that both $(A \times B) \times C$ and $A \times (B \times C)$ satisfy this universal property. Deduce that $(A \times B) \times C$ and $A \times (B \times C)$ are necessarily isomorphic.

Solution. In any category C in which binary products are defined, there exists an object $P \coloneqq \prod_{i=1}^{3} A_i$ and morphisms $(\pi_i : P \to A_i)_1^3$ so that for any object $Z \in \mathsf{Obj}(\mathsf{C})$ and family of morphisms $(\varphi_i : Z \to A_i)_1^3$ there exists a uniquely defined morphism $\varphi : Z \to P$ so that the diagram below commutes for every $1 \leqslant i \leqslant 3$.

$$\begin{tikzcd}
& Z \arrow[dl, "\varphi_i"'] \arrow[d, "\varphi"] \\
A_i & P \arrow[l, "\pi_i"]
\end{tikzcd}$$

There are three such commutative diagrams for $i = 1, 2, 3$ where $A_1 = A, A_2 = B, A_3 = C$. The commutativity relations are as follows

$$\pi_i \circ \varphi = \varphi_i, \qquad 1 \leqslant i \leqslant 3.$$

We will only show that $(A_1 \times A_2) \times A_3$ satisfies this universal property. Indeed, notice that there are projections $\pi' : (A_1 \times A_2) \times A_3 \to (A_1 \times A_2)$ and $\pi_3 : (A_1 \times A_2) \times A_3 \to A_3$ Then, suppose Z is any other object with morphisms $(\varphi_i : Z \to A_i)_1^3$. Next, notice that there is a unique morhpism $\varphi_1 \times \varphi_2 : Z \to A_1 \times A_2$ by the universal property of the product $A_1 \times A_2$ which satisfies

$$\pi_1' \circ (\varphi_1 \times \varphi_2) = \varphi_1,$$

$$\pi_2' \circ (\varphi_1 \times \varphi_2) = \varphi_2$$

where $\pi_1' : (A_1 \times A_2) \to A_1$ and $\pi_2' : (A_1 \times A_2) \to A_2$ are the canonical projections of the product $A_1 \times A_2$ to its factors A_1 and A_2 respectively.

Next, due the the universal property of the binary product $(A_1 \times A_2) \times A_3$, there is a unique morphism $(\varphi_1 \times \varphi_2) \times \varphi_3 : Z \to (A \times B) \times C$ so that

$$\begin{aligned}
\pi' \circ ((\varphi_1 \times \varphi_2) \times \varphi_3) =& \varphi_1 \times \varphi_2, \\
\pi_3 \circ ((\varphi_1 \times \varphi_2) \times \varphi_3) =& \varphi_3
\end{aligned}$$

We notice that $(A_1 \times A_2) \times A_3$ has projections to A_1 as $\pi_1 \coloneqq \pi_1' \circ \pi'$, to A_2 as $\pi_2 \coloneqq \pi_2' \circ \pi'$ and to A_3 as π_3. This tells us that

$$\begin{aligned}
\pi_1 \circ ((\varphi_1 \times \varphi_2) \times \varphi_3) =& \pi_1' \circ (\varphi_1 \times \varphi_2) \\
=& \varphi_1
\end{aligned}$$

and

$$\pi_2 \circ ((\varphi_1 \times \varphi_2) \times \varphi_3) = \pi'_2 \circ (\varphi_1 \times \varphi_2)$$
$$= \varphi_2$$

and the final equality is

$$\pi_3 \circ ((\varphi_1 \times \varphi_2) \times \varphi_3) = \varphi_3$$

We simply define our φ to be this morphism $(\varphi_1 \times \varphi_2) \times \varphi_3$ and clearly from the above proof it is uniquely defined.

This completes the proof that $(A_1 \times A_2) \times A_3$ also satisfies the universal property of products of triplets.

For uniqueness, suppose Z with the morphisms $(\rho_i : Z \to A_i)_1^3$ also satisfies the same universal property. Then, by universal property of P there is a unique morphism $\varphi : Z \to P$ that satisfies $\pi_i \circ \varphi = \rho_i$ for every $1 \leqslant i \leqslant 3$. Again, by universal property of Z there is a unique morphism $\varphi' : P \to Z$ so that $\rho_i \circ \varphi' = \pi_i$. We claim that $\varphi \circ \varphi' = 1_P$ and $\varphi' \circ \varphi = 1_Z$. Notice that $\pi_i \circ (\varphi \circ \varphi') = \rho_i \circ \varphi' = \pi_i$. Again, by universal property of P we find that 1_P is the unique morphism which satisfies $\pi_i \circ 1_P = \pi_i$for every $1 \leqslant i3$. This proves that $\varphi \circ \varphi' = 1_P$. Similarly, by invoking universal property of Z, we find that 1_Z is the unique morphism satisfiying $\rho_i \circ 1_Z = \rho_i$ but this means that it must be identical to the morphism $\varphi' \circ \varphi$ which also satisfies $\rho_i \circ (\varphi' \circ \varphi) = \pi_i \circ \varphi = \rho_i$. This completes the proof that P is unique up to isomorphism. ■

Exercise. [Aluffi, Algebra, Ch-1, Ex-5.10] Push the enveloope a little further still, and define products and coproducts for families (i.e., indexed sets) of objects of a category.

Do these exist in Set.

Solution. Suppose we are given a family of objects $(A_i)_{i \in I}$ in a category C. An object $P = \prod_I A_i$ is said to be a product of the A_i together with a family of morphisms $(\pi_i : P \to A_i)_{i \in I}$ if it satisfies the following universal property:

Given any object $Z \in \mathsf{Obj}(\mathsf{C})$ and a family of morphisms $(\rho_i : Z \to A_i)_{i \in I}$ there exists a unique morphism $\varphi : Z \to P$ such that $\pi_i \circ \varphi = \rho_i$ for every $i \in I$. Such a P is uniquely determined up to isomorphisms.

I claim that given a family of sets $(A_i)_{i \in I}$, their generalized cartesian product

$$\prod_{i \in I} A_i \coloneqq \left\{ f : I \to \bigcup_{i \in I} A_i \mid f(i) \in A_i \text{ for every } i \in I \right\}$$

is the product. Notice there are projections $\pi_i : \prod_{i\in I} A_i \to A_i$ for every $i \in I$ so that $\pi_i(f) = f(i)$. Given any object C with a family of functions $(\rho_i : C \to A_i)_{i\in I}$ there exists a uniquely defined function $\varphi : C \to \prod_{i\in I} A_i$ such that for every $i \in I$, $\pi_i \circ \varphi = \rho_i$. Indeed, simply define $\varphi(c) = f_c$ so that $f_c(i) = \rho_i(c)$. That φ is well defined as a function and satisfies the commutativity relation for every $i \in I$ is obvious. We are left to show that φ is uniquely defined. For this, suppose φ' is another well defined map $C \to \prod_{i\in I} A_i$ so that $\pi_i \circ \varphi' = \rho_i$.

We will show that $\varphi = \varphi'$. For this, take $c \in C$, then for all $i \in I$ we find that $(\varphi'(c))(i) = \pi_i(\varphi'(c)) = \rho_i(c) = \pi_i(\varphi(c)) = (\varphi(c))(i)$ and this completes the proof of uniqueness of φ. ∎

Exercise. [Aluffi, Algebra, Ch-1, Ex-5.11] Already did in a solution to some secion 1.1/1.2 problem.

Exercise. [Aluffi, Algebra, Ch-1, Ex-5.12] Define the notions of *fibered products* and *fibered coproducts*, as terminal objects of the categories $\mathsf{C}_{\alpha,\beta}, \mathsf{C}^{\alpha,\beta}$ considered in Eg-3.10 by stating carefully the corresponding universal properties.

As it happens, Set has both fibered products and fibered coproducts. Define those objects 'concretely', in terms of naive set theory.

Solution. TODO

References

[Aluffi, Algebra]